



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis II (MAT321)

General Metric Space

Properties of Metric, Modified Metric, NLS, IPS

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CONDUCTED BY

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Motivation of Metric, Norm and Inner Product

Metric Space

Metric Space

A **metric space** is a pair (X, d) where $X \neq \emptyset$ and $d : X \times X \rightarrow \mathbb{R}$ satisfies, for all $x, y, z \in X$:

$$(M1) \quad d(x, y) \geq 0;$$

$$(M2) \quad d(x, y) = 0 \iff x = y;$$

$$(M3) \quad d(x, y) = d(y, x);$$

$$(M4) \quad d(x, y) \leq d(x, z) + d(z, y).$$

Here d is called a **metric** on X .

Pseudo-Metric Space

A **pseudo-metric space** is a pair (X, d) where $X \neq \emptyset$ and $d : X \times X \rightarrow \mathbb{R}$ satisfies, for all $x, y, z \in X$:

$$(PM1) \quad d(x, y) \geq 0;$$

$$(PM2) \quad d(x, y) = d(y, x);$$

$$(PM3) \quad d(x, z) \leq d(x, y) + d(y, z).$$

Here d is called a **pseudo-metric** on X .

Metric on Real Line

💡 Euclidean Metric on Real Line

Show that $d : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ defined as

$$d(x, y) = |x - y|$$

is a metric on $\mathbb{R}$.



Another Metric on Real Line

Show that $d : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ defined as

$$d(x, y) = 3|x - y|$$

is a metric on $\mathbb{R}$.



🔍 Problem

Show that $d : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ defined as

$$d(x, y) = |x - 2y|$$

is a not metric on $\mathbb{R}$.



Problem

Show that $d : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ defined as

$$d(x, y) = (x - y)^2$$

is a not metric on $\mathbb{R}$.



Inequalities in Metric Spaces

Reverse Triangle Inequality

Let (X, d) be a metric space. Show that for all $x, y, z \in X$,

$$|d(x, z) - d(y, z)| \leq d(x, y).$$



Two-point inequality for distances

Let (X, d) be a metric space. Show that for all $x, y, z, w \in X$,



$$|d(x, y) - d(z, w)| \leq d(x, z) + d(y, w).$$

Discrete Metric Space

💡 Discrete Metric

Let X be a non-empty set. Define $d : X \times X \rightarrow \mathbb{R}$ as

$$d(x, y) = \begin{cases} 0, & \text{if } x = y; \\ 1, & \text{if } x \neq y. \end{cases}$$

Show that d is a metric on X .



Modified Metrics

💡 Metric via a Fixed Point

Let (X, d) be a metric space and let $x_0 \in X$. Define a function $\delta : X \times X \rightarrow \mathbb{R}$ by

$$\delta(x, y) = \begin{cases} d(x, x_0) + d(x_0, y), & \text{if } x \neq y, \\ 0, & \text{if } x = y. \end{cases}$$

Show whether δ is a metric on X or not.

Bounded Transform of a Metric

Let (X, d) be a metric space. Define $d' : X \times X \rightarrow \mathbb{R}$ as

$$d'(x, y) = \frac{d(x, y)}{1 + d(x, y)}.$$

Show that d' is a metric on X .



Metric via Truncation

Let (X, d) be a metric space. Define $d' : X \times X \rightarrow \mathbb{R}$ as

$$d'(x, y) = \min\{1, d(x, y)\}.$$

Show that d' is a metric on X .



Pullback Metric along an Injection

💡 Pullback metric along an Injection

Let (Y, d) be a metric space and let $\alpha : X \rightarrow Y$ be an injection. Define $d' : X \times X \rightarrow \mathbb{R}$ as

$$d'(x, y) = d(\alpha(x), \alpha(y)).$$

Show that d' is a metric on X .

Isometric Spaces

Isometric Spaces

Let (X, d_X) and (Y, d_Y) be metric spaces. They are said to be **isometric** if there exists a bijection $\alpha : X \rightarrow Y$ such that for all $x_1, x_2 \in X$,

$$d_Y(\alpha(x_1), \alpha(x_2)) = d_X(x_1, x_2).$$

Here α is called an **isometry** from X to Y .

Motivation of Norm

Normed Linear Space

Normed Linear Space

A **normed linear space** is a pair $(X, \|\cdot\|)$ where X is a vector space over $\mathbb{R}$ (or $\mathbb{C}$) and $\|\cdot\| : X \rightarrow \mathbb{R}$ satisfies, for all $x, y \in X$ and $\alpha \in \mathbb{R}$ (or $\mathbb{C}$):

$$(N1) \quad \|x\| \geq 0;$$

$$(N2) \quad \|x\| = 0 \iff x = 0;$$

$$(N3) \quad \|\alpha x\| = |\alpha| \|x\|;$$

$$(N4) \quad \|x + y\| \leq \|x\| + \|y\|.$$

Here $\|\cdot\|$ is called a **norm** on X .

Metric Induced by a Norm

Let $(X, \|\cdot\|)$ be a normed space. Define $d : X \times X \rightarrow \mathbb{R}$ as

$$d(x, y) = \|x - y\|.$$

Show that d is a metric on X .

Motivation of Inner Product

Inner Product Space

Inner Product Space

An **inner product space** is a pair $(X, \langle \cdot, \cdot \rangle)$ where X is a vector space over $\mathbb{R}$ (or $\mathbb{C}$) and $\langle \cdot, \cdot \rangle : X \times X \rightarrow \mathbb{R}$ (or $\mathbb{C}$) satisfies, for all $x, y, z \in X$ and $\alpha \in \mathbb{R}$ (or $\mathbb{C}$):

$$(IP1) \quad \langle x, x \rangle \geq 0;$$

$$(IP2) \quad \langle x, x \rangle = 0 \iff x = 0;$$

$$(IP3) \quad \langle x, y \rangle = \overline{\langle y, x \rangle};$$

$$(IP4) \quad \langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle.$$

Here $\langle \cdot, \cdot \rangle$ is called an **inner product** on X .

💡 Norm Induced by an Inner Product

Let $(X, \langle \cdot, \cdot \rangle)$ be an inner product space. Define $\| \cdot \| : X \rightarrow \mathbb{R}$ as 

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

Show that $\| \cdot \|$ is a norm on X .

Metric Induced by an Inner Product

Let $(X, \langle \cdot, \cdot \rangle)$ be an inner product space. Define $d : X \times X \rightarrow \mathbb{R}$ as 

$$d(x, y) = \|x - y\| = \sqrt{\langle x - y, x - y \rangle}.$$

Show that d is a metric on X .

💡 Cauchy-Schwarz Inequality

Let $(X, \langle \cdot, \cdot \rangle)$ be an inner product space. Show that for all $x, y \in X$,

$$|\langle x, y \rangle| \leq \sqrt{\langle x, x \rangle} \sqrt{\langle y, y \rangle}.$$

or equivalently,

$$|\langle x, y \rangle| \leq \|x\| \|y\|.$$

Thank You!

We'd love your questions and feedback.

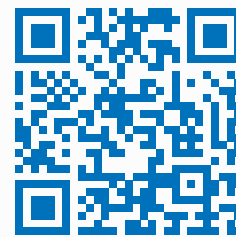
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(Lectures, walkthroughs, and course updates)



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References

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- [2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.